

– SAFETY DATA SHEET

NN-AluBond High Alumina Binder

High-Alumina Refractory Binder (Calcium Aluminate Cement)

Prepared in the format of GHS / the Chemicals (MSIHC) Rules

Document No.	Revision	Issue Date	Product Code
NNPL/SDS/PROD-09	1.0	2 July 2026	NN-AluBond

01 IDENTIFICATION

Field	Detail
Product name	NN-AluBond High Alumina Binder
Product type	Calcium aluminate cement (high-alumina refractory binder)
CAS number	65997-16-2 (aluminous / calcium aluminate cement)
Recommended use	Hydraulic binder for refractory castables, mortars and rapid-strength concretes
Supplier	NilayNarayan Polychem LLP, Plot D-167 (P), Kandra Ind. Area, Govindpur, Jharkhand 826010
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02 HAZARD IDENTIFICATION

GHS classification: Skin Irritation Category 2 (H315); Serious Eye Damage Category 1 (H318); STOT SE Category 3 – respiratory irritation (H335). Wetted cement is alkaline.

- **Eyes** – H318 – Causes serious eye damage.
- **Skin** – H315 – Causes skin irritation; wet cement may cause burns/dermatitis (alkaline).
- **Dust** – H335 – Dust may cause respiratory irritation.

Precautionary: avoid dust; wear eye protection, gloves and a dust mask; wash promptly after skin contact; keep out of reach of children.

03 COMPOSITION / INFORMATION ON INGREDIENTS

Ingredient	CAS No.	Conc.	Remark
Calcium aluminate cement	65997-16-2	≥ 95 %	Hydraulic binder (calcium aluminates)
Calcium aluminate phases (CaO·Al ₂ O ₃ etc.)	—	Main	Reactive hydraulic phase
Residual crystalline silica	14808-60-7	Trace	Keep exposure low

04 FIRST-AID MEASURES

- **Inhalation** – Move to fresh air. If dust inhaled and irritation persists, seek medical advice.
- **Skin** – Wash skin with plenty of water; remove contaminated clothing. For wet-cement burns or persistent irritation, seek medical attention.
- **Eyes** – Rinse immediately with plenty of water for at least 15 minutes; remove contact lenses if easy; seek urgent medical attention.

- **Ingestion** — Rinse mouth; do not induce vomiting; drink water; obtain medical attention.

05 FIRE-FIGHTING MEASURES

- **Media** — Non-combustible and non-flammable. Use media appropriate to the surrounding fire.
- **Hazards** — No special decomposition hazards; product is inorganic and stable.
- **Protection** — Wear dust protection when handling disturbed powder near a fire.

06 ACCIDENTAL RELEASE MEASURES

- **Precautions** — Avoid dust generation; wear PPE. Do not dry-sweep vigorously — use vacuum or damp methods.
- **Containment** — Collect dry material into containers for reuse or disposal; avoid contact with water before intended use.
- **Environment** — Prevent large releases into drains and watercourses (alkaline; may raise pH).

07 HANDLING & STORAGE

- **Handling** — Avoid dust and skin/eye contact; wear PPE; maintain good hygiene; do not eat or drink while handling.
- **Storage** — Store dry, cool and covered; keep bags sealed and off the floor; highly moisture-sensitive.
- **Segregation** — Keep away from acids and from Portland cement/lime except in controlled formulations.

08 EXPOSURE CONTROLS / PERSONAL PROTECTION

Control	Recommendation
Cement / nuisance dust	Observe local OEL for inert/nuisance dust and respirable crystalline silica
Respiratory protection	Dust mask (P2) where dust is generated
Hand protection	Waterproof / nitrile gloves (wet cement is alkaline)
Eye protection	Safety goggles
Engineering controls	Dust extraction / ventilation; eyewash facility

09 PHYSICAL & CHEMICAL PROPERTIES

Property	Value
Appearance	Fine powder, grey to buff / off-white
Odour	Odourless
pH (in water)	Alkaline ($\approx 11 - 12$ when wetted)
Fineness (Blaine)	$\geq 300 \text{ m}^2/\text{kg}$
Solubility in water	Reacts (hydrates) rather than dissolves
Flammability	Non-combustible
Bulk density	$\approx 1000 - 1300 \text{ kg/m}^3$ (loose powder)

10 STABILITY & REACTIVITY

- **Stability** — Stable when kept dry; hydrates and hardens on contact with water (exothermic).

- **Avoid** — Moisture / humidity (pre-hydration and loss of activity); contact with water before use.
- **Incompatibles** — Acids (react, effervesce); mixing with Portland cement / lime can cause flash set.
- **Decomposition** — None significant; inorganic and thermally stable.

11 TOXICOLOGICAL INFORMATION

- **Irritation** — Irritant to eyes and skin; wet cement is alkaline and may cause burns/dermatitis on prolonged contact.
- **Respiratory** — Dust may irritate the nose, throat and respiratory tract; limit exposure to any crystalline silica.
- **Systemic** — Low acute systemic toxicity; effects are principally local (irritation).

12 ECOLOGICAL INFORMATION

Not toxic in the classical sense; large releases raise the pH of water and soil. Prevent uncontrolled entry into drains and watercourses. Cured material is inert.

13 DISPOSAL CONSIDERATIONS

Allow surplus to hydrate/set where safe and dispose of the inert solid per local regulation, or dispose of dry powder through authorised handlers. Do not discharge powder or slurry to drains.

14 TRANSPORT INFORMATION

Not classified as dangerous goods for transport (inorganic hydraulic cement). No UN number assigned. Keep dry during transit; confirm against current transport regulations before shipment.

15 REGULATORY INFORMATION

High-alumina refractory binder supplied to the applicable Indian standards (IS 15895 for high alumina refractory cement; IS 6452 for structural high alumina cement). Handle per applicable chemical-safety and occupational-dust regulations.

16 OTHER INFORMATION

This SDS is provided for safe-handling guidance. Classification and exposure limits must be verified against the regulations in force at the place of use. Issued by the NilayNarayan Polychem LLP NABL-accredited laboratory.

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